

M21_202604

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Setup

Load libraries

1. Set `ggplot2` parameters

1 Load Files

```
# Load data: rt words

rt_w <- read_csv("rt_words_frq.csv") %>%
  filter(correct == 1) %>%
  mutate(
    SubjID = factor(SubjID),
    Item = factor(target),
    logRT = log(response_time),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
```

```

    zLogFS = as.numeric(scale(LogFS))
  )

# Load data: rt nonwords
rt_nw <- read_csv("rt_nwords_frq.csv") %>%
  filter(correct == 1) %>%
  mutate(
    SubjID = factor(SubjID),
    ItemID = factor(ItemID),
    complexity = factor(complexity,
      levels = c("SIMP", "COMP"),
      labels = c("Simple", "Complex")),
    logRT = log(response_time),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

# Load data: erp_fs words
erp_w <- read_csv("erp_fs_dmg_words.csv") %>%
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_fs nonwords
erp_nw <- read_csv("erp_fs_dmg_nonwords.csv") %>%
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size, levels = c("Small", "Large")),
    complexity = factor(complexity,
      labels = c("Simple", "Complex")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

```

2 Response Time (RT)

- **Base Frequency** is your primary lexical familiarity index for **RT models** — it drives the behavioral effects and is interpretable as overall lexical availability. Family Size can be included as a covariate in RT models but should not be foregrounded.

2.1 Words — RT

Question: Does orthographic sensitivity predict efficiency, controlling lexical familiarity?

Theoretical framing: RTs index processing speed and overall efficiency. The finding that OS predicts overall RT but does not modulate the structure of morphological effects is theoretically meaningful — it suggests that orthographic skill affects how fast the system operates without changing what it does. This is consistent with discriminative learning accounts in which OS indexes the precision of sublexical cue weights, which would speed all form-based processing rather than selectively facilitating morphologically structured items.

RT models: Strip out the morphological structure × individual difference interactions entirely. RTs answer Question 1 only — who is faster — and the models should be specified accordingly. Running leaner models also strengthens the RT results by removing the fragile interaction terms that are currently cluttering them. Interactions between morphological variables and individual differences are Question 2 and 3 territory. RT models establish the efficiency baseline cleanly, without prejudging what the ERP models will show.

```
library(lmerTest)
```

```

m_rt_words <- lmer(
  logRT ~ zLogBF + zLogFS + OS + LP +
    (1 | SubjID) +
    (1 | Item),
  data = rt_w,
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_rt_words)

```

```

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: logRT ~ zLogBF + zLogFS + OS + LP + (1 | SubjID) + (1 | Item)
|| Data: rt_w
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: -3536.3

```

```

||
|| Scaled residuals:
||   Min       1Q   Median       3Q      Max
|| -9.8884 -0.6626 -0.0643  0.5951  5.1184
||
|| Random effects:
||   Groups   Name                Variance Std.Dev.
||   Item      (Intercept) 0.001807 0.04251
||   SubjID     (Intercept) 0.017718 0.13311
||   Residual                    0.029621 0.17211
|| Number of obs: 5847, groups: Item, 100; SubjID, 67
||
|| Fixed effects:
||              Estimate Std. Error        df t value Pr(>|t|)
|| (Intercept)   6.390069    0.016965   72.434536 376.655 < 2e-16 ***
|| zLogBF        -0.017184    0.004696  100.974079  -3.659 0.000404 ***
|| zLogFS        -0.012858    0.004865   93.591020  -2.643 0.009629 **
|| OS            -0.034712    0.018734   63.842814  -1.853 0.068518 .
|| LP             0.011591    0.012386   63.973484   0.936 0.352893
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) zLogBF zLogFS OS
|| zLogBF      0.009
|| zLogFS      0.002 -0.105
|| OS           0.012 -0.001  0.001
|| LP           0.008  0.000  0.001  0.008
||
|| # Effect sizes for individual terms
|| eta_squared(m_rt_words, partial = TRUE)
||
|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) |      95% CI
|| -----|-----|-----|
|| zLogBF    |          0.12 | [0.04, 1.00]
|| zLogFS    |          0.07 | [0.01, 1.00]
|| OS        |          0.05 | [0.00, 1.00]
|| LP        |          0.01 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
||
|| # Model R²
|| r2_nakagawa(m_rt_words)
||
|| # R2 for Mixed Models
||
|| Conditional R2: 0.417
|| Marginal R2: 0.033
||
|| • Base Frequency as the primary lexical familiarity predictor
|| • Family Size included as a covariate but not interacted or foregrounded

```

What to report:

- Main effect of zLogBF — this is your primary finding, should be significant and negative (higher frequency = faster RT). Report and interpret.
- Main effect of zLogFS — this is a covariate. If significant, note it briefly. If not, report the null without dwelling on it.
- Main effect of OS — report whether it reaches significance or shows a trend. This is theoretically meaningful either way: a significant effect supports the efficiency claim; a null effect should be noted honestly.
- Main effect of LP — report as null. One sentence is sufficient.

What not to report in text: The random effects structure, the optimizer setting, or any discussion of model convergence unless there was a problem.

2.2 Nonwords — RT

```

library(lmerTest)

contrasts(rt_nw$complexity) <- c(-0.5, 0.5)

m_rt_nw <- lmer(
  logRT ~ complexity + zLogBF + zLogFS + OS + LP +
    (1 + complexity | SubjID) +
    (1 | ItemID),
  data = rt_nw,
  control = lmerControl(optimizer = "bobyqa")
)
m_rt_nw

```

```

|| Linear mixed model fit by REML ['lmerModLmerTest']
|| Formula: logRT ~ complexity + zLogBF + zLogFS + OS + LP + (1 + complexity | SubjID) + (1 | ItemID)
|| Data: rt_nw
|| REML criterion at convergence: -4165.782
|| Random effects:
|| Groups Name Std.Dev. Corr
|| ItemID (Intercept) 0.03662
|| SubjID (Intercept) 0.13143
|| complexity1 0.02365 -0.17
|| Residual 0.14641
|| Number of obs: 4607, groups: ItemID, 100; SubjID, 67
|| Fixed Effects:
|| (Intercept) complexity1 zLogBF zLogFS OS LP
|| 6.554486 0.048807 0.015469 -0.006359 -0.041063 0.003398
summary(m_rt_nw)

```

```

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: logRT ~ complexity + zLogBF + zLogFS + OS + LP + (1 + complexity | SubjID) + (1 | ItemID)
|| Data: rt_nw
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: -4165.8
||
|| Scaled residuals:
|| Min 1Q Median 3Q Max
|| -4.2605 -0.6625 -0.0227 0.6633 3.8775
||
|| Random effects:
|| Groups Name Variance Std.Dev. Corr
|| ItemID (Intercept) 0.0013408 0.03662
|| SubjID (Intercept) 0.0172747 0.13143
|| complexity1 0.0005594 0.02365 -0.17
|| Residual 0.0214360 0.14641
|| Number of obs: 4607, groups: ItemID, 100; SubjID, 67
||
|| Fixed effects:
|| Estimate Std. Error df t value Pr(>|t|)
|| (Intercept) 6.554486 0.016639 68.832580 393.923 < 2e-16 ***
|| complexity1 0.048807 0.005347 62.675281 9.128 4.07e-13 ***
|| zLogBF 0.015469 0.005080 98.214341 3.045 0.00299 **
|| zLogFS -0.006359 0.005137 96.636227 -1.238 0.21880
|| OS -0.041063 0.018454 62.212180 -2.225 0.02970 *
|| LP 0.003398 0.012228 62.906744 0.278 0.78204
|| ---
|| Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
|| (Intr) cmplx1 zLogBF zLogFS OS
|| complexity1 -0.073
|| zLogBF -0.004 -0.005
|| zLogFS -0.004 0.011 -0.576
|| OS 0.011 -0.007 -0.001 -0.001
|| LP 0.003 -0.015 -0.002 0.000 0.009
# Effect sizes for individual terms
eta_squared(m_rt_nw, partial = TRUE)

```

```

|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) | 95% CI
|| -----|-----|-----
|| complexity | 0.57 | [0.44, 1.00]
|| zLogBF | 0.09 | [0.02, 1.00]
|| zLogFS | 0.02 | [0.00, 1.00]
|| OS | 0.07 | [0.00, 1.00]
|| LP | 1.23e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
# Model R²
r2_nakagawa(m_rt_nw)

```

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.494
|| Marginal R2: 0.050

```

- Complexity as the primary morphological contrast
- Base Frequency as lexical familiarity covariate

What to report:

- Main effect of Complexity — your primary finding, should be large and significant (complex slower than simple). Report and interpret in terms of the processing cost of morphological structure without a stored whole-word representation.
- Main effect of zLogBF — report the inhibitory effect (higher frequency = slower rejection). This is theoretically interesting and worth a sentence of interpretation: high-frequency stems generate strong cue activation that competes with the rejection decision, consistent with discriminative learning accounts.
- Main effect of zLogFS — covariate. Report briefly whether significant or null.
- Main effect of OS — report. If significant, it speaks to overall efficiency for novel forms. If null, note it.
- Main effect of LP — report as null. One sentence.

What not to report in text: Again, do not narrate the absence of interactions you did not include — do not say “we did not find a Complexity × OS interaction” because that interaction is not in this model by design. If a reviewer asks, you explain it was a Question 2 test reserved for the ERP.

```
library(emmeans)

# Step 1: Get emmeans on log scale with Satterthwaite df
emm_log <- emmeans(m_rt_nw, ~ complexity, lmer.df = "kenward-roger", pbkrtest.limit = 5000)
summary(emm_log)

|| complexity emmean      SE    df lower.CL upper.CL
|| Simple      6.529 0.0172 70.03   6.495   6.563
|| Complex     6.578 0.0168 70.15   6.544   6.611
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95

# Step 2: Back-transform manually
emm_summary <- summary(emm_log)
emm_summary$mean_ms <- exp(emm_summary$emmean)
# emm_summary$lower_ms <- exp(emm_summary$asympt.LCL) # Uncomment for asymptotic df calculation
# emm_summary$upper_ms <- exp(emm_summary$asympt.UCL)
emm_summary$lower_ms <- exp(emm_summary$lower.CL) # Uncomment for kenward-roger df calculation
emm_summary$upper_ms <- exp(emm_summary$upper.CL)
emm_summary

|| complexity emmean      SE    df lower.CL upper.CL mean_ms lower_ms upper_ms
|| Simple      6.529 0.0172 70.03   6.495   6.563   684.6   661.5   708.5
|| Complex     6.578 0.0168 70.15   6.544   6.611   718.9   695.2   743.4
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95

diff_ms <- exp(emm_summary$emmean[emm_summary$complexity == "Complex"]) -
  exp(emm_summary$emmean[emm_summary$complexity == "Simple"])
diff_ms

|| [1] 34.24345
```

3 ERP N250

Restructure so that OS is the primary moderator and LP is a covariate in the words analysis, while both are tested for nonwords but with OS foregrounded. The three-way Complexity × Family Size × OS interaction for nonwords should be dropped unless you have a specific theoretical reason to retain it — it is currently one of the results that is hardest to interpret and easiest for reviewers to challenge.

3.1 Words — 250

Question: Does orthographic sensitivity modulate early sensitivity to morphological family structure?

```
# Set contrast on the data column, not a model object
contrasts(erp_w$family_size) <- c(-0.5, 0.5)

m_w_N250 <- lmer(
  value ~ family_size * OS + LP +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N250)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ family_size * OS + LP + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_w, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 15026.8
||
|| Scaled residuals:
```

```

||      Min      1Q  Median      3Q      Max
|| -8.4669 -0.4045 -0.0218  0.3824 12.9276
||
|| Random effects:
|| Groups      Name      Variance Std.Dev.
|| SubjID:chlabel (Intercept) 4.625   2.151
|| SubjID      (Intercept) 2.918   1.708
|| Residual                2.754   1.659
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    -0.75979    0.22921   57.00000   -3.315   0.0016 **
|| family_size1   -0.32357    0.05837 1617.99999   -5.543 3.46e-08 ***
|| OS              0.26085    0.25635   57.00000    1.018   0.3132
|| LP             -0.27860    0.18636   57.00000   -1.495   0.1404
|| family_size1:OS -0.11120    0.06532 1617.99999   -1.702   0.0889 .
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) fmly_1 OS      LP
|| family_size1  0.000
|| OS           -0.046  0.000
|| LP            0.038  0.000  0.021
|| fmly_sz1:OS   0.000 -0.047  0.000  0.000

```

Effect sizes for individual terms
`eta_squared(m_w_N250, partial = TRUE)`

```

|| # Effect Size for ANOVA (Type III)
||
|| Parameter      | Eta2 (partial) |      95% CI
|| -----|-----|-----|
|| family_size    |      0.02      | [0.01, 1.00]
|| OS              |      0.02      | [0.00, 1.00]
|| LP              |      0.04      | [0.00, 1.00]
|| family_size:OS | 1.79e-03      | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```

Model R²
`r2_nakagawa(m_w_N250)`

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.738
|| Marginal R2: 0.019

```

3.1.1 Main Effect of Family Size

```

library(emmeans)
emmeans(m_w_N250, ~ family_size)

|| family_size emmean SE df lower.CL upper.CL
|| Large      -0.571 0.231 58.9   -1.03   -0.110
|| Small      -0.900 0.231 58.9   -1.36   -0.438
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95

```

3.2 Nonwords — N250

```

# Set contrasts on data columns before the model
contrasts(erp_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_nw$family_size) <- c(-0.5, 0.5)

m_nw_N250 <- lmer(
  value ~ complexity * OS + complexity * LP + family_size +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_nw, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_nw_N250)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ complexity * OS + complexity * LP + family_size + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_nw, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")

```

```

||
|| REML criterion at convergence: 29933.2
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -7.7366 -0.5107 -0.0056  0.4938 12.2676
||
|| Random effects:
||  Groups             Name             Variance Std.Dev.
||  SubjID:chlabel      (Intercept)    3.109      1.763
||  SubjID              (Intercept)    2.163      1.471
||  Residual                        4.072      2.018
|| Number of obs: 6480, groups:  SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)   -0.66509    0.19682   56.99997  -3.379 0.001317 **
|| complexity1   -0.23263    0.05023 4856.00002  -4.631 3.73e-06 ***
|| OS              0.29808    0.22013   56.99997   1.354 0.181051
|| LP             -0.20675    0.16003   56.99997  -1.292 0.201577
|| family_size1   0.18172    0.05014 4856.00002   3.625 0.000292 ***
|| complexity1:OS  0.39199    0.05618 4856.00002   6.978 3.40e-12 ***
|| complexity1:LP  0.35764    0.04084 4856.00002   8.758 < 2e-16 ***
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) cmplx1 OS      LP      fmly_1 cm1:OS
|| complexity1  0.000
|| OS           -0.046 0.000
|| LP            0.038 0.000 0.021
|| family_siz1  0.000 0.000 0.000 0.000
|| cmplxty1:OS  0.000 -0.046 0.000 0.000 0.000
|| cmplxty1:LP  0.000 0.038 0.000 0.000 0.000 0.021

```

3.2.1 Main Effect of Complexity

Simple means for each level — establishes direction and magnitude:

```

library(emmeans)
emm_options(lmer.df = "kenward-roger")

emm_complexity <- emmeans(m_nw_N250, ~ complexity)
summary(emm_complexity)

|| complexity emmean    SE df asymp.LCL asymp.UCL
|| Simple     -0.526 0.198 Inf   -0.914   -0.138
|| Complex    -0.759 0.198 Inf   -1.148   -0.371
||
|| Results are averaged over the levels of: family_size
|| Degrees-of-freedom method: asymptotic
|| Confidence level used: 0.95

pairs(emm_complexity)

|| contrast      estimate      SE df z.ratio p.value
|| Simple - Complex  0.234 0.0501 Inf   4.657 <.0001
||
|| Results are averaged over the levels of: family_size
|| Degrees-of-freedom method: asymptotic

```

3.2.2 Main Effect of Family Size

```

emm_fs <- emmeans(m_nw_N250, ~ family_size)
summary(emm_fs)

|| family_size emmean    SE df asymp.LCL asymp.UCL
|| Small       -0.734 0.198 Inf   -1.12   -0.345
|| Large       -0.552 0.198 Inf   -0.94   -0.164
||
|| Results are averaged over the levels of: complexity
|| Degrees-of-freedom method: asymptotic
|| Confidence level used: 0.95

pairs(emm_fs)

|| contrast      estimate      SE df z.ratio p.value
|| Small - Large  -0.182 0.0501 Inf   -3.625 0.0003
||
|| Results are averaged over the levels of: complexity
|| Degrees-of-freedom method: asymptotic

```

3.2.3 Complexity $\times$ OS Interaction

This is your primary Question 2 test. You want to know how the complexity effect changes across levels of OS. The standard approach is to probe OS at -1, 0, and +1 SD:

```
# Complexity effect at three levels of OS
emm_complexity_OS <- emmeans(m_nw_N250,
                             ~ complexity | OS,
                             at = list(OS = c(-1, 0, 1)))
summary(emm_complexity_OS)

|| OS = -1:
|| complexity emmean    SE df asymp.LCL asymp.UCL
|| Simple    -0.632 0.304 Inf   -1.229   -0.0356
|| Complex   -1.274 0.304 Inf   -1.871   -0.6775
||
|| OS = 0:
|| complexity emmean    SE df asymp.LCL asymp.UCL
|| Simple    -0.530 0.198 Inf   -0.919   -0.1416
|| Complex   -0.780 0.198 Inf   -1.169   -0.3915
||
|| OS = 1:
|| complexity emmean    SE df asymp.LCL asymp.UCL
|| Simple    -0.428 0.291 Inf   -0.998    0.1415
|| Complex   -0.286 0.291 Inf   -0.856    0.2836
||
|| Results are averaged over the levels of: family_size
|| Degrees-of-freedom method: asymptotic
|| Confidence level used: 0.95

# Pairwise complexity contrast at each level of OS
pairs(emm_complexity_OS)

|| OS = -1:
|| contrast      estimate    SE df z.ratio p.value
|| Simple - Complex    0.642 0.0771 Inf   8.329 <.0001
||
|| OS = 0:
|| contrast      estimate    SE df z.ratio p.value
|| Simple - Complex    0.250 0.0502 Inf   4.979 <.0001
||
|| OS = 1:
|| contrast      estimate    SE df z.ratio p.value
|| Simple - Complex   -0.142 0.0736 Inf  -1.932 0.0534
||
|| Results are averaged over the levels of: family_size
|| Degrees-of-freedom method: asymptotic

# Test whether the complexity contrast changes across OS levels
# (this is the formal test of the interaction slope)
# emtrends(m_nw_N250, ~ OS, var = "complexity")
```

3.2.4 Complexity $\times$ LP Interaction. Parallel follow-up for LP:

```
# Complexity effect at three levels of LP
emm_complexity_LP <- emmeans(m_nw_N250,
                             ~ complexity | LP,
                             at = list(LP = c(-1, 0, 1)))
summary(emm_complexity_LP)

|| LP = -1:
|| complexity emmean    SE df asymp.LCL asymp.UCL
|| Simple    -0.159 0.251 Inf   -0.650    0.332
|| Complex   -0.733 0.251 Inf   -1.224   -0.242
||
|| LP = 0:
|| complexity emmean    SE df asymp.LCL asymp.UCL
|| Simple    -0.545 0.198 Inf   -0.933   -0.156
|| Complex   -0.761 0.198 Inf   -1.149   -0.372
||
|| LP = 1:
|| complexity emmean    SE df asymp.LCL asymp.UCL
|| Simple    -0.930 0.260 Inf   -1.441   -0.420
|| Complex   -0.789 0.260 Inf   -1.299   -0.278
||
|| Results are averaged over the levels of: family_size
|| Degrees-of-freedom method: asymptotic
|| Confidence level used: 0.95

# Pairwise complexity contrast at each level of LP
pairs(emm_complexity_LP)
```



```

|| LP = -1:
|| contrast      estimate      SE df z.ratio p.value
|| Simple - Complex    0.574 0.0634 Inf  9.047 <.0001
||
|| LP = 0:
|| contrast      estimate      SE df z.ratio p.value
|| Simple - Complex    0.216 0.0502 Inf  4.310 <.0001
||
|| LP = 1:
|| contrast      estimate      SE df z.ratio p.value
|| Simple - Complex   -0.141 0.0659 Inf -2.145 0.0320
||
|| Results are averaged over the levels of: family_size
|| Degrees-of-freedom method: asymptotic
# Formal test of interaction slope
# emtrends(m_nw_N250, ~ LP, var = "complexity1")

```

3.2.5 Direct Comparison of the Two Interaction Slopes

This is theoretically important — you want to know whether the Complexity $\times$ OS and Complexity $\times$ LP interactions are meaningfully different from each other, since your theoretical framing predicts both should be present at the N250 but potentially with different magnitudes.

The slopes themselves — $\beta = 0.392$ for OS and $\beta = 0.358$ for LP — are very similar in magnitude, which is consistent with your finding that both dimensions modulate early parsing efficiency for nonwords. Reporting both slopes alongside the follow-up contrasts at ± 1 SD will allow you to make this point clearly in the write-up.

3.2.5.1 What to Report From Each

Test	What to report
Complexity main effect	Means $\pm$ CIs for simple and complex, pairwise contrast
Family Size main effect	Means $\pm$ CIs for large and small, pairwise contrast
Complexity $\times$ OS	Complexity contrast at -1, 0, +1 SD of OS; note where effect reverses or disappears
Complexity $\times$ LP	Complexity contrast at -1, 0, +1 SD of LP; same logic
Slope comparison	Both emtrends values to show similarity of the two interaction slopes

The pattern you are looking for in both interactions is the same: at **low skill** the complexity effect should be large and positive (*complex more positive than simple*), at mean skill attenuated, and at **high skill** *reduced or reversed* — consistent with your existing results showing that higher-skill readers show more efficient early parsing of novel morphological structure.

4 ERP: N400

Question: Does language proficiency shape semantic integration of morphological information (N400)?

Restructure so that LP is the primary moderator and OS is a covariate. The three-way Complexity $\times$ Family Size $\times$ LP for nonwords is your strongest finding and should be the centerpiece of Question 3 — but it needs to be the planned primary test rather than one of several equally-weighted interactions, which is how it currently reads.

Theoretical framing: The N400 indexes semantic integration. The prediction is that LP — which indexes depth and breadth of lexical-semantic knowledge — should modulate the N400 response to morphological family structure, particularly for nonwords where semantic co-activation from family members must be generated from stem-level cues alone. OS is included as a covariate. The key finding is the three-way Complexity $\times$ Family Size $\times$ LP interaction for nonwords, which is your strongest evidence for the predicted role of LP in morphological family activation during semantic integration.

4.1 N400 — Words

```
contrasts(erp_w$family_size) <- c(-0.5, 0.5)
```

```

m_w_N400 <- lmer(
  value ~ family_size * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N400)

```

```

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ family_size * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_w, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 15561.8
||

```

```

|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -10.9307  -0.3874   0.0081   0.4071   6.2048
||
|| Random effects:
|| Groups      Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept)  5.238      2.289
|| SubjID       (Intercept)  4.575      2.139
|| Residual                3.304      1.818
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    2.618e-01  2.842e-01  5.700e+01   0.921   0.3608
|| family_size1   -4.522e-01  6.392e-02  1.618e+03  -7.074  2.23e-12 ***
|| LP              1.231e-03  2.311e-01  5.700e+01   0.005   0.9958
|| OS              6.771e-01  3.179e-01  5.700e+01   2.130   0.0375 *
|| family_size1:LP  7.114e-02  5.201e-02  1.618e+03   1.368   0.1716
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) fmly_1 LP      OS
|| family_size1  0.000
|| LP            0.038  0.000
|| OS            -0.046  0.000  0.021
|| fmly_sz1:LP   0.000  0.039  0.000  0.000

```

4.1.1 Main Effect of Family Size

```

emm_fs_w_N400 <- emmeans(m_w_N400, ~ family_size)
summary(emm_fs_w_N400)

```

```

|| family_size emmean    SE    df lower.CL upper.CL
|| Large        0.5179  0.286  58.5   -0.0535    1.089
|| Small        0.0623  0.286  58.5   -0.5091    0.634
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95
pairs(emm_fs_w_N400)

```

```

|| contrast      estimate    SE    df t.ratio p.value
|| Large - Small    0.456  0.0639  1618    7.134  <.0001
||
|| Degrees-of-freedom method: kenward-roger

```

4.2 N400 — Nonwords

Question: Does language proficiency determine how familiarity modulates complexity costs?

```

contrasts(erp_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_nw$family_size) <- c(-0.5, 0.5)

```

```

m_nw_N400 <- lmer(
  value ~ complexity * family_size * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_nw, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_nw_N400)

```

```

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ complexity * family_size * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_nw, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 32979.6
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -13.9873  -0.4837  -0.0118   0.4655  14.1595
||
|| Random effects:
|| Groups      Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept)  6.389      2.528
|| SubjID       (Intercept)  5.568      2.360

```

```

|| Residual                6.114    2.473
|| Number of obs: 6480, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    -0.45163    0.31311   57.00002   -1.442 0.154666
|| complexity1    -0.45956    0.06148  4853.99999   -7.475 9.11e-14 ***
|| family_size1    0.08743    0.06148  4853.99999    1.422 0.155086
|| LP              0.01139    0.25458   57.00002    0.045 0.964463
|| OS              0.38123    0.35019   57.00002    1.089 0.280904
|| complexity1:family_size1 -0.44008    0.12296  4853.99999  -3.579 0.000348 ***
|| complexity1:LP    0.25043    0.05003  4853.99999    5.006 5.77e-07 ***
|| family_size1:LP    0.12120    0.05003  4853.99999    2.423 0.015450 *
|| complexity1:family_size1:LP 0.22717    0.10006  4853.99999    2.270 0.023230 *
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) cmplx1 fmly_1 LP      OS      cm1:_1 cm1:LP f_1:LP
|| complexity1  0.000
|| family_size1 0.000 0.000
|| LP           0.038 0.000 0.000
|| OS           -0.046 0.000 0.000 0.021
|| cmplxty1:_1  0.000 0.000 0.000 0.000 0.000
|| cmplxty1:LP  0.000 0.039 0.000 0.000 0.000 0.000
|| fmly_sz1:LP  0.000 0.000 0.039 0.000 0.000 0.000 0.000
|| cmpl1:_1:LP 0.000 0.000 0.000 0.000 0.000 0.039 0.000 0.000

```

4.2.1 Complexity × Family Size × LP interaction

The three-way needs to be unpacked by examining the Complexity × Family Size interaction separately at each level of LP. This tells you how the joint influence of morphological complexity and family size on semantic integration changes as a function of language proficiency — which is exactly your theoretical claim.

```

# Step 1: Unpack the three-way
# Complexity x Family Size at three levels of LP
emm_3way <- emmeans(m_nw_N400,
  ~ complexity * family_size | LP,
  at = list(LP = c(-1, 0, 1)))
summary(emm_3way)

```

```

|| LP = -1:
|| complexity family_size emmean SE df asymp.LCL asymp.UCL
|| Simple Small -0.2420 0.401 Inf -1.028 0.5442
|| Complex Small -0.6184 0.401 Inf -1.405 0.1678
|| Simple Large 0.0578 0.401 Inf -0.728 0.8441
|| Complex Large -0.9858 0.401 Inf -1.772 -0.1995
||
|| LP = 0:
|| complexity family_size emmean SE df asymp.LCL asymp.UCL
|| Simple Small -0.3596 0.317 Inf -0.982 0.2622
|| Complex Small -0.5992 0.317 Inf -1.221 0.0227
|| Simple Large -0.0522 0.317 Inf -0.674 0.5697
|| Complex Large -0.7318 0.317 Inf -1.354 -0.1099
||
|| LP = 1:
|| complexity family_size emmean SE df asymp.LCL asymp.UCL
|| Simple Small -0.4773 0.417 Inf -1.294 0.3398
|| Complex Small -0.5799 0.417 Inf -1.397 0.2371
|| Simple Large -0.1622 0.417 Inf -0.979 0.6549
|| Complex Large -0.4778 0.417 Inf -1.295 0.3393
||
|| Degrees-of-freedom method: asymptotic
|| Confidence level used: 0.95

```

```

# Step 2: Pairwise complexity contrasts within each
# family size condition at each level of LP
pairs(emm_3way, simple = "complexity")

```

```

|| family_size = Small, LP = -1:
|| contrast estimate SE df z.ratio p.value
|| Simple - Complex 0.376 0.1100 Inf 3.424 0.0006
||
|| family_size = Large, LP = -1:
|| contrast estimate SE df z.ratio p.value
|| Simple - Complex 1.044 0.1100 Inf 9.494 <.0001
||
|| family_size = Small, LP = 0:
|| contrast estimate SE df z.ratio p.value
|| Simple - Complex 0.240 0.0869 Inf 2.755 0.0059
||

```

```

|| family_size = Large, LP = 0:
|| contrast      estimate      SE df z.ratio p.value
|| Simple - Complex    0.680 0.0869 Inf   7.816 <.0001
||
|| family_size = Small, LP = 1:
|| contrast      estimate      SE df z.ratio p.value
|| Simple - Complex    0.103 0.1140 Inf   0.899 0.3687
||
|| family_size = Large, LP = 1:
|| contrast      estimate      SE df z.ratio p.value
|| Simple - Complex    0.316 0.1140 Inf   2.763 0.0057
||
|| Degrees-of-freedom method: asymptotic
# Step 3: Pairwise family size contrasts within each
# complexity condition at each level of LP
pairs(emm_3way, simple = "family_size")

```

```

|| complexity = Simple, LP = -1:
|| contrast      estimate      SE df z.ratio p.value
|| Small - Large   -0.300 0.1100 Inf  -2.728 0.0064
||
|| complexity = Complex, LP = -1:
|| contrast      estimate      SE df z.ratio p.value
|| Small - Large    0.367 0.1100 Inf   3.342 0.0008
||
|| complexity = Simple, LP = 0:
|| contrast      estimate      SE df z.ratio p.value
|| Small - Large   -0.307 0.0869 Inf  -3.536 0.0004
||
|| complexity = Complex, LP = 0:
|| contrast      estimate      SE df z.ratio p.value
|| Small - Large    0.133 0.0869 Inf   1.525 0.1272
||
|| complexity = Simple, LP = 1:
|| contrast      estimate      SE df z.ratio p.value
|| Small - Large   -0.315 0.1140 Inf  -2.758 0.0058
||
|| complexity = Complex, LP = 1:
|| contrast      estimate      SE df z.ratio p.value
|| Small - Large   -0.102 0.1140 Inf  -0.894 0.3711
||
|| Degrees-of-freedom method: asymptotic

```

```

# Step 4: The interaction contrast - does the complexity x
# family size interaction itself change across LP levels?
# This is the formal test of the three-way
emm_2way_by_LP <- emmeans(m_nw_N400,
  ~ complexity * family_size | LP,
  at = list(LP = c(-1, 0, 1)))
contrast(emm_2way_by_LP,
  interaction = c("pairwise", "pairwise"))

```

```

|| LP = -1:
|| complexity_pairwise family_size_pairwise estimate      SE df z.ratio p.value
|| Simple - Complex    Small - Large             -0.667 0.155 Inf  -4.292 <.0001
||
|| LP = 0:
|| complexity_pairwise family_size_pairwise estimate      SE df z.ratio p.value
|| Simple - Complex    Small - Large             -0.440 0.123 Inf  -3.579 0.0003
||
|| LP = 1:
|| complexity_pairwise family_size_pairwise estimate      SE df z.ratio p.value
|| Simple - Complex    Small - Large             -0.213 0.162 Inf  -1.318 0.1875
||
|| Degrees-of-freedom method: asymptotic

```

4.2.2 Complexity × Family Size interaction averaged across LP

```

# Complexity x Family Size averaged over LP
emm_2way <- emmeans(m_nw_N400, ~ complexity * family_size)
summary(emm_2way)

```

```

|| complexity family_size emmean      SE df asymp.LCL asymp.UCL
|| Simple      Small      -0.3540 0.317 Inf    -0.975    0.2674
|| Complex     Small      -0.6001 0.317 Inf    -1.221    0.0213
|| Simple      Large      -0.0469 0.317 Inf    -0.668    0.5745
|| Complex     Large      -0.7440 0.317 Inf    -1.365   -0.1227
||
|| Degrees-of-freedom method: asymptotic
|| Confidence level used: 0.95

```

```

pairs(emm_2way, simple = "complexity")

|| family_size = Small:
|| contrast      estimate      SE  df z.ratio p.value
|| Simple - Complex    0.246 0.0869 Inf   2.833 0.0046
||
|| family_size = Large:
|| contrast      estimate      SE  df z.ratio p.value
|| Simple - Complex    0.697 0.0869 Inf   8.025 <.0001
||
|| Degrees-of-freedom method: asymptotic
pairs(emm_2way, simple = "family_size")

|| complexity = Simple:
|| contrast      estimate      SE  df z.ratio p.value
|| Small - Large   -0.307 0.0869 Inf  -3.535 0.0004
||
|| complexity = Complex:
|| contrast      estimate      SE  df z.ratio p.value
|| Small - Large    0.144 0.0869 Inf   1.657 0.0975
||
|| Degrees-of-freedom method: asymptotic

```

4.2.3 Complexity × LP interaction averaged over Family Size:

```

# Complexity x LP at three levels of LP
emm_complexity_LP_N400 <- emmeans(m_nw_N400,
  ~ complexity | LP,
  at = list(LP = c(-1, 0, 1)))
summary(emm_complexity_LP_N400)

|| LP = -1:
|| complexity  emmean      SE  df asymp.LCL asymp.UCL
|| Simple     -0.0921 0.397  Inf   -0.871   0.6867
|| Complex    -0.8021 0.397  Inf   -1.581  -0.0233
||
|| LP = 0:
|| complexity  emmean      SE  df asymp.LCL asymp.UCL
|| Simple     -0.2059 0.314  Inf   -0.822   0.4101
|| Complex    -0.6655 0.314  Inf   -1.281  -0.0495
||
|| LP = 1:
|| complexity  emmean      SE  df asymp.LCL asymp.UCL
|| Simple     -0.3197 0.413  Inf   -1.129   0.4896
|| Complex    -0.5289 0.413  Inf   -1.338   0.2805
||
|| Results are averaged over the levels of: family_size
|| Degrees-of-freedom method: asymptotic
|| Confidence level used: 0.95
pairs(emm_complexity_LP_N400)

|| LP = -1:
|| contrast      estimate      SE  df z.ratio p.value
|| Simple - Complex    0.710 0.0777 Inf   9.135 <.0001
||
|| LP = 0:
|| contrast      estimate      SE  df z.ratio p.value
|| Simple - Complex    0.460 0.0615 Inf   7.475 <.0001
||
|| LP = 1:
|| contrast      estimate      SE  df z.ratio p.value
|| Simple - Complex    0.209 0.0808 Inf   2.589 0.0096
||
|| Results are averaged over the levels of: family_size
|| Degrees-of-freedom method: asymptotic

```

4.2.3.1 N400 (nonwords): (Complex - Simple) difference scores as a function of continuous Language Proficiency (LP), averaged over Family Size

```

# Generate complexity differences across LP
lp_grid <- seq(-2, 2, by = 0.25)

emm_lp <- emmeans(
  m_nw_N400,
  ~ complexity | LP,
  at = list(LP = lp_grid),
  weights = "proportional"
)

```

```

)

# Compute Complex - Simple difference at each LP
diff_lp <- contrast(emm_lp, method = "pairwise")
df_diff_lp <- as.data.frame(summary(diff_lp, infer = c(TRUE, TRUE)))

# Plot difference scores vs LP
ggplot(df_diff_lp, aes(x = LP, y = estimate)) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_line(size = 1) +
  geom_ribbon(
    aes(ymin = asymp.LCL, ymax = asymp.UCL),
    alpha = 0.25
  ) +
  labs(
    x = "Language Proficiency (LP; SD units)",
    y = "N400 Complexity Effect (Complex - Simple,  $\mu$ V)",
    title = "N400 Nonwords: Morphological Complexity Effect as a Function of Language Proficiency",
    subtitle = "Model-estimated difference scores averaged over family size"
  ) +
  theme_classic()

```

